

Explaining by ~~Removing~~: A Unified Framework for Model Explanation*

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Motivation

There is a need for a unifying framework

Minimal Image Representation approach, Zhou et al. (2014) — LIME (Ribeiro et al., 2016)

Masking Model approach (Dabkowski and Gal, 2017) — SHAP (Lundberg and Lee, 2017)

Prediction Difference Analysis (Zintgraf et al., 2017)

Meaningful Perturbations (Fong and Vedaldi, 2017) — RISE (Petsiuk et al., 2018)

?

FIDO-CA (Chang et al., 2018) — INVASE (Yoon et al., 2018)

Extremal Perturbations (Fong et al., 2019) — REAL-X (Jethani et al., 2021a)

...

Contributions

1. A unified framework that characterizes **26** existing explanation methods → **removal-based explanations**.
2. **Mathematical tools** to represent different approaches for **removing features** from ML models.
3. Removal-based explanations are implicitly tied to **cooperative game theory** → advantages of the **Shapley value** over alternative allocation strategies.
4. Feature removal is a simple application of **subtractive counterfactual reasoning**.

Previous Unifying Efforts

2017

A Unified Approach to Interpreting Model Predictions

2017

**TOWARDS BETTER UNDERSTANDING OF
GRADIENT-BASED ATTRIBUTION METHODS
FOR DEEP NEURAL NETWORKS**

Marco Ancona
Department of Computer Science
ETH Zurich, Switzerland

Enea Ceolini
Institute of Neuroinformatics
University Zürich and ETH Zürich

2020

**Understanding Global Feature Contributions With
Additive Importance Measures**

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LIME, DeepLIFT, LRP,
QII, ...



SHAP

Grad * Input, DeepLIFT, LRP and Integrated
Gradients



modified gradient back
propagations

permutation tests, Shapley Net Effects,
feature ablation, SAGE...



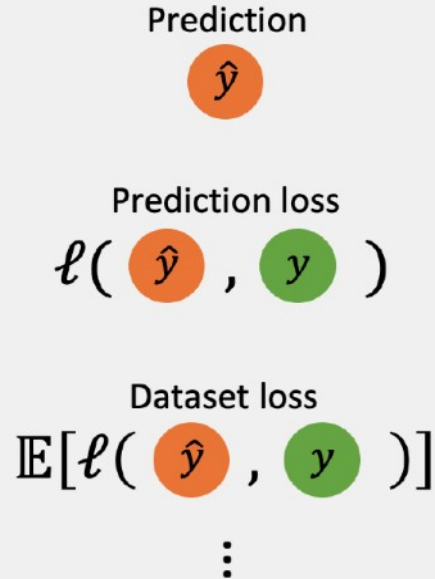
additive importance measures

The Removal-based Explanations Framework

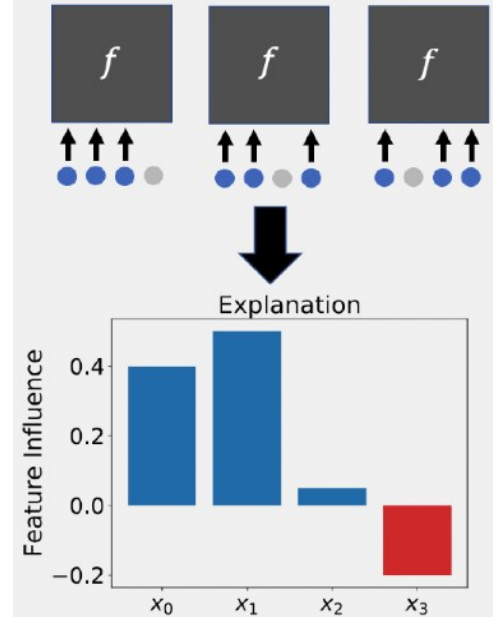
1. Feature removal



2. Model behavior



3. Summary technique



Methods Survey

Key insights

1. Common
 - a. marginalize out removed features with their conditional distribution
 - b. Shapley values
2. Relatively unique and spatially isolated
 - a. RISE; LIME for tabular data; INVASE
3. Many combinations left unexplored!

		Summary technique						
		Feature attribution			Feature selection			
		Remove individual	Include individual	Mean when included	Shapley value	Additive model	High value subset	Low value subset
Feature removal	Zeros	Occlusion CXPlain		RISE				MM
	Default values					LIME (images)		
	Extend pixels						MIR	
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	Generative model						FIDO-CA	
	Marginalize (replacement distribution)					LIME (tabular)		
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	Marginalize (conditional)	PredDiff Conditional perm. test			SHAP SAGE LossSHAP Shapley Effects			
	Tree distribution				TreeSHAP			
	Surrogate model						LZX REAL-X	
	Missingness during training						INVASE	
	Separate models	Feature ablation	Univariate predictors		Shapley Net Effects IME 2009 SPVIM			

Model behavior

■ Prediction ■ Prediction loss ■ Prediction mean loss ■ Dataset loss ■ Prediction loss (output) ■ Dataset loss (output)

Mathematical Formulation

Overview

1. Feature removal

$$F: \mathcal{X} \times \mathcal{P}(D) \mapsto \mathcal{Y}$$

2. Model behavior

$$u: \mathcal{P}(D) \mapsto \mathbb{R}$$

3. Summary technique

$$E: U \mapsto \mathbb{R}^d$$

or

$$E: U \mapsto \mathcal{P}(D)$$

$$F: \mathcal{X} \times \mathcal{P}(D) \mapsto \mathcal{Y}$$

Defining Feature Removal

$$F(x) = f(\text{features_to_keep}, \text{features_to_remove})$$

- Zero ablation
- Default values
- Generative model
- Train separate models
 - Train surrogate models
- “Missingness” during training

$$F(x_S) = f(x_S, 0).$$

$$F(x_S) = f(x_S, r_{\bar{S}}).$$

$$F(x_S) = f(x_S, \tilde{x}_{\bar{S}}).$$

$$F: \mathcal{X} \times \mathcal{P}(D) \mapsto \mathcal{Y}$$

Defining Feature Removal (cont)

Marginalization

- Marginalize with conditional

- Tree distribution

$$F(x_S) = \mathbb{E}[f(X) \mid X_S = x_S].$$

- Marginalize with marginal

$$F(x_S) = \mathbb{E}[f(x_S, X_{\bar{S}})].$$

- Marginalize with product of marginals

$$F(x_S) = \mathbb{E}_{\prod_{i \in D} p(X_i)}[f(x_S, X_{\bar{S}})].$$

- Marginalize with uniform

$$F(x_S) = \mathbb{E}_{\prod_{i \in D} u_i(X_i)}[f(x_S, X_{\bar{S}})].$$

- Marginalize with replacement distribution

$$F(x, S) = \mathbb{E}_{\prod_{i \in D} q_{x_i}(X_i)}[f(x_S, X_{\bar{S}})].$$

$$u: \mathcal{P}(D) \mapsto \mathbb{R}$$

Explaining Model Behaviors

- Given the newly defined model $F(x_S)$, we need a metric to assess how important the x_S features are
- Options
 - Prediction*
 - Prediction loss*
 - Prediction mean loss
 - Dataset loss*
 - Prediction loss w.r.t. output
 - Dataset loss w.r.t. output

$$F(x_S).$$

$$-\ell(F(x_S), y).$$

$$-\mathbb{E}_{p(Y|X=x)}[\ell(F(x_S), Y)].$$

$$-\mathbb{E}_{XY}[\ell(F(X_S), Y)].$$

$$-\ell(F(x_S), F(x)).$$

$$-\mathbb{E}_X[\ell(F(X_S), F(X))].$$

$$E: \mathcal{U} \mapsto \mathbb{R}^d$$

or

$$E: \mathcal{U} \mapsto \mathcal{P}(D)$$

Summarizing Feature Influence

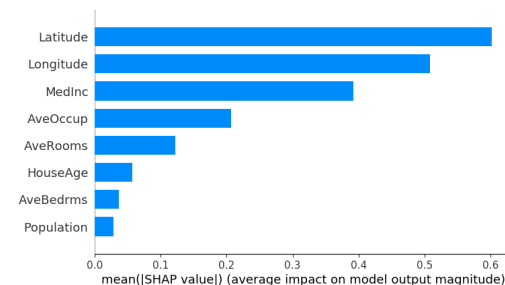
Two related approaches

- Map feature to real number value (feature attribution)*
- Map dataset to a set of important features (feature selection)

$$E: \mathcal{U} \rightarrow \mathbb{R}^d$$

$$E: \mathcal{U} \rightarrow \mathcal{P}(D)$$

Both are related, often the second is just a threshold applied to the first



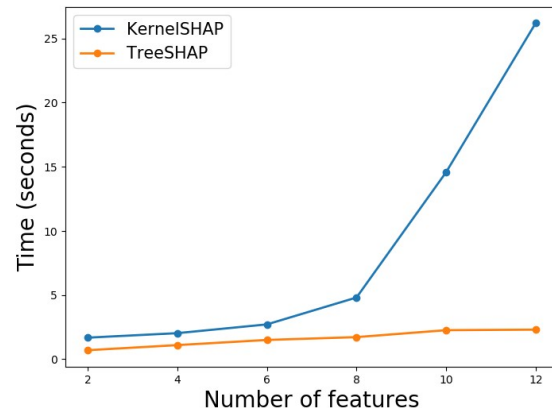
Computational Complexity

How to isolate features?

- Consider every subset including the feature in question
 - Exact solutions are $O(2^d / d)$ in the worst case
 - SHAP, RISE summarization, LIME additive model, etc
- Only consider the subset where you remove the feature
 - Polynomial in d
 - Occlusion, PRedDiff, CXPlain, permutation tests, etc

Approximations? (much faster, but lose worst-case guarantees)

- Sampling
- TreeSHAP (dynamic programming solution for tree models)
- Solve a continuous relaxation of the subset problem (i.e. learn a mask)
- Greedy search (MIR)
- Learn a model to do the explanation task (map from dataset to feature importance vector)



Connections to Other Theoretical Frameworks

Game Theory

Same setup as what the SHAP people said

- Cooperative games where we solve for allocations

Makes SHAP unique; however, other methods can be viewed as fitting (linear, additive) models to cooperative games

- Proofs in appendix, not critical

SUMMARIZATION	METHODS	RELATED TO
Shapley value	Shapley Net Effects, IME, QII, SHAP, TreeSHAP, KernelSHAP, LossSHAP, Shapley Effects, SAGE, SPVIM	Shapley value, probabilistic values, modeling cooperative games
Mean value when included	RISE	Banzhaf value, probabilistic values, modeling cooperative games
Remove/include individual players	Occlusion, PredDiff, CXPlain, permutation tests, univariate predictors, feature ablation (LOCO)	Probabilistic values, modeling cooperative games
Fit additive model	LIME	Shapley value, Banzhaf value, modeling cooperative games
High/low value coalitions	MP, EP, MIR, MM, L2X, INVASE, REAL-X, FIDO-CA	Maximum/minimum excess

Information Theory

- f approximates response variable's conditional distribution
 - Classification: $f(x) \approx p(Y|X = x)$
 - Regression: $f(x) \approx \mathbb{E}[Y|X = x]$

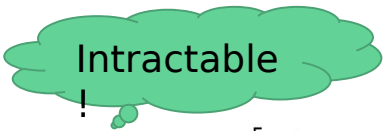
Information Theory

- f approximates response variable's conditional distribution $f(x) \approx p(Y|X = x)$
 - Classification: $f(x) \approx \mathbb{E}[Y|X = x]$
 - Regression:
- F can also approximate conditional distribution (over subsets!) $\{q(Y|X_S) : S \subseteq D\}$
 - Set of conditional distributions:

Information Theory

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 - Classification: $f(x) \approx \mathbb{E}[Y|X = x]$
 - Regression:
- F can also approximate conditional distribution (over subsets!) $\{q(Y|X_S) : S \subseteq D\}$
 - Set of conditional $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$:
- Want this to be probabilistically valid (called **consistent**)
 - Countable Additivity:
 - Bayes' Rule:
 - Occurs only when average over conditional distribution

Approximations of SHAP



Intractable

Conditional Distribution is Intractable $\mathbb{E}[f(X) \mid X_S = x_S] = \mathbb{E}_{p(X_{\bar{S}} \mid X_S = x_S)}[f(x_S, X_{\bar{S}})]$

- Assume feature independence
- Assume model linearity

$$\approx \mathbb{E}_{p(X_{\bar{S}})}[f(x_S, X_{\bar{S}})]$$

$$\approx f(x_S, \mathbb{E}[X_{\bar{S}}])$$

Approximations of SHAP

Intractable

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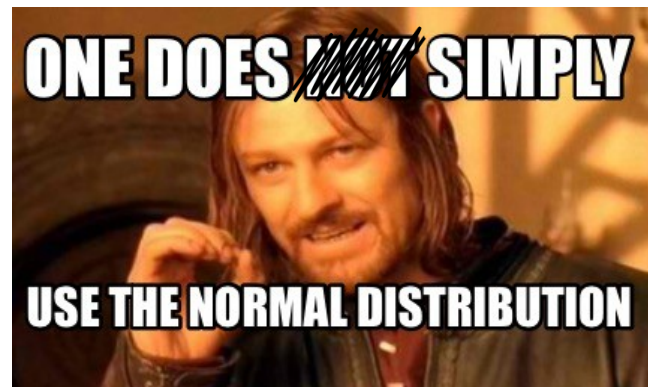
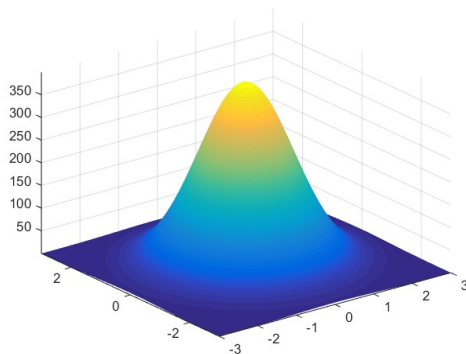
- Assume feature independence
- Assume model linearity

$$\approx \mathbb{E}_{p(X_{\bar{S}})}[f(x_S, X_{\bar{S}})]$$

$$\approx f(x_S, \mathbb{E}[X_{\bar{S}}])$$

Or:

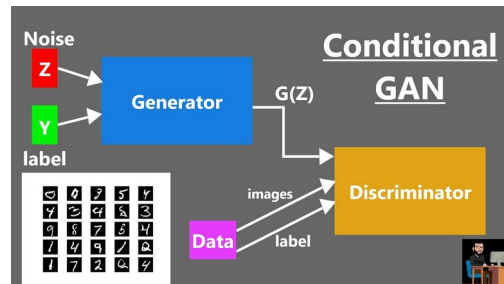
- Assume normal



Approximations of SHAP

- Generative Model

- Draw samples from cGAN
- Single-Sample Monte Carlo Approximation



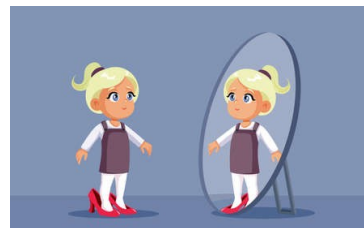
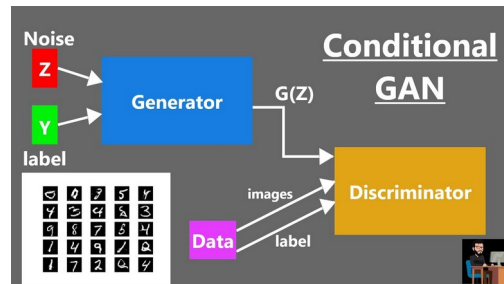
Approximations of SHAP

- Generative Model

- Draw samples from cGAN
- Single-Sample Monte Carlo Approximation

- Surrogate Model

- Train a model to match model predictions
- Objective $\min_F \mathbb{E}_X \mathbb{E}_S [\ell(F(X_S), f(X))]$



Approximations of SHAP

● Generative Model

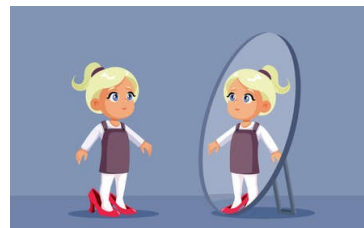
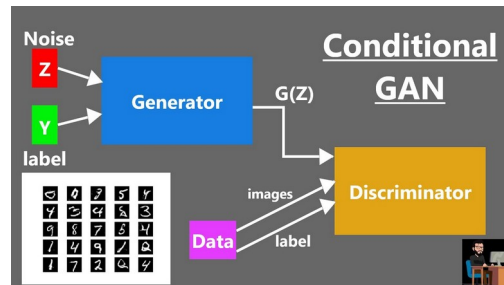
- Draw samples from cGAN
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● Surrogate Model

- Train a model to match model predictions
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● Training with Missing

- Train your original model with missing features
- Objective $\min_F \mathbb{E}_{XY} \mathbb{E}_S [\ell(F(X_S), Y)]$



Approximations of SHAP

● Generative Model

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● Surrogate Model

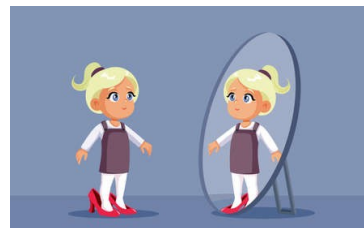
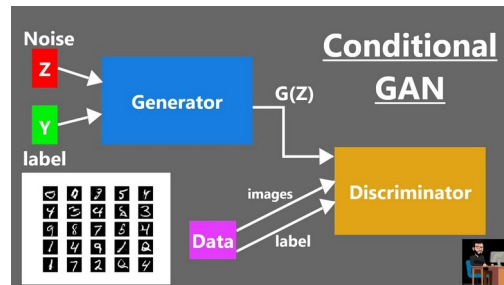
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● Training with Missing

- Train your original model with missing features
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● Separate Model

- Train separate model for each possible subset



Information Theory Quantities

MODEL BEHAVIOR	SET FUNCTION	METHODS	RELATED TO
Prediction	u_x	Occlusion, MIR, MM, IME, QII, LIME, MP, EP, FIDO-CA, RISE, SHAP, KernelSHAP, TreeSHAP	Conditional probability, conditional expectation
Prediction loss	v_{xy}	LossSHAP, CXPlain	Pointwise mutual information
Prediction mean loss	v_x	INVASE	KL divergence with conditional distribution
Dataset loss	v	Permutation tests, univariate predictors, feature ablation (LOCO), Shapley Net Effects, SAGE, SPVIM	Mutual information (with label)
Prediction loss (output)	w_x	L2X, REAL-X	KL divergence with full model output
Dataset loss (output)	w	Shapley Effects	Mutual information (with output)

Cognition Theory

- Subtractive Counterfactual (Epstude and Roese, 2008) and Method of Difference (Mill, 1884)
- Norm Theory and the downhill rule
- Trade-off between simplicity and completeness



Experiments

Summary technique

		Feature attribution				Feature selection			
		Remove individual	Include individual	Mean when included	Shapley value	Additive model	High value subset	Low value subset	Partitioned subsets
Feature removal	Zeros	Occlusion CXPlain		RISE					MM
	Default values					LIME (images)			
	Extend pixels						MIR		
	Blurring						EP	MP	
	Generative model						FIDO-CA		
	Marginalize (replacement distribution)					LIME (tabular)			
	Marginalize (uniform)				IME 2010				
	Marginalize (marginals product)				QII				
	Marginalize (marginal)	Permutation test			SHAP KernelSHAP				
	Marginalize (conditional)	PredDiff Conditional perm. test			SHAP SAGE LossSHAP Shapley Effects				
	Tree distribution				TreeSHAP				
	Surrogate model						L2X REAL-X		
	Missingness during training						INVASE		
	Separate models	Feature ablation	Univariate predictors		Shapley Net Effects IME 2009 SPVIM				

Model behavior

Prediction

Prediction loss

Prediction mean loss

Dataset loss

Prediction loss (output)

Dataset loss (output)

Model behavior

■ Prediction
■ Prediction loss
■ Prediction mean loss
■ Dataset loss
■ Prediction loss (output)
■ Dataset loss (output)

Summary technique

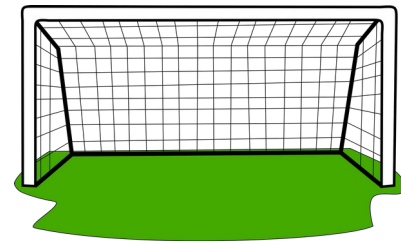
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Separate models	Feature ablation	Univariate predictors		Shapley Net Effects IME 2009 SPVIM				

So many unexplored combinations!!!

Model behavior

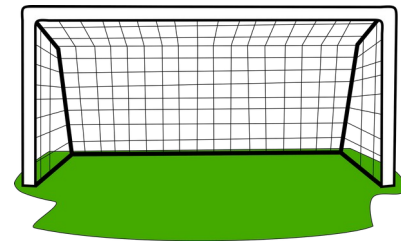
■ Prediction
 ■ Prediction loss
 ■ Prediction mean loss
 ■ Dataset loss
 ■ Prediction loss (output)
 ■ Dataset loss (output)

Experimental Goals



1. Fill in the gaps of removal-based method combinations

Experimental Goals



1. Fill in the gaps of removal-based method combinations

SHAP!

2. Verify that the information-theoretic model works best

3. Verify other theorized relationships between existing methods

Removal-Based Model Combinations

Feature Removal

- Default Values
- Marginalization
 - Uniform
 - Product
 - Joint

Model Behavior

- Prediction
- Prediction Loss
- Dataset Loss

Summary

Technique

- Removing
- Including
- Mean when Included
- Banzhaf
- Shapley

Removal-Based Model Combinations

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Model Behavior

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Summary

Technique

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- Mean when Included
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- Shapley



Over 80
Combination
s Tested!

Removal-Based Model Combinations

Feature Removal

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Model Behavior

- Prediction
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Summary

Technique

- Removing
- Including
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Experimental Domains

- Census Income

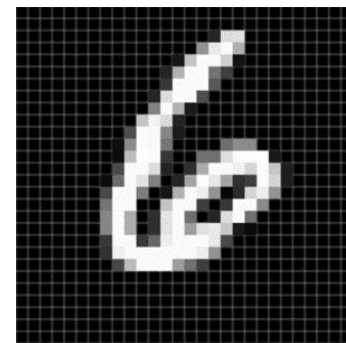
- 48,842 Individuals
- 14 Socioeconomic Features, >\$50k income or not
- Light-GBM (gradient boosted tree)

- MNIST

- 70,000 Handwritten Digits
- 32x32 Grayscale Pixels
- CNN (14 Layers)

- Breast Cancer Subtypes

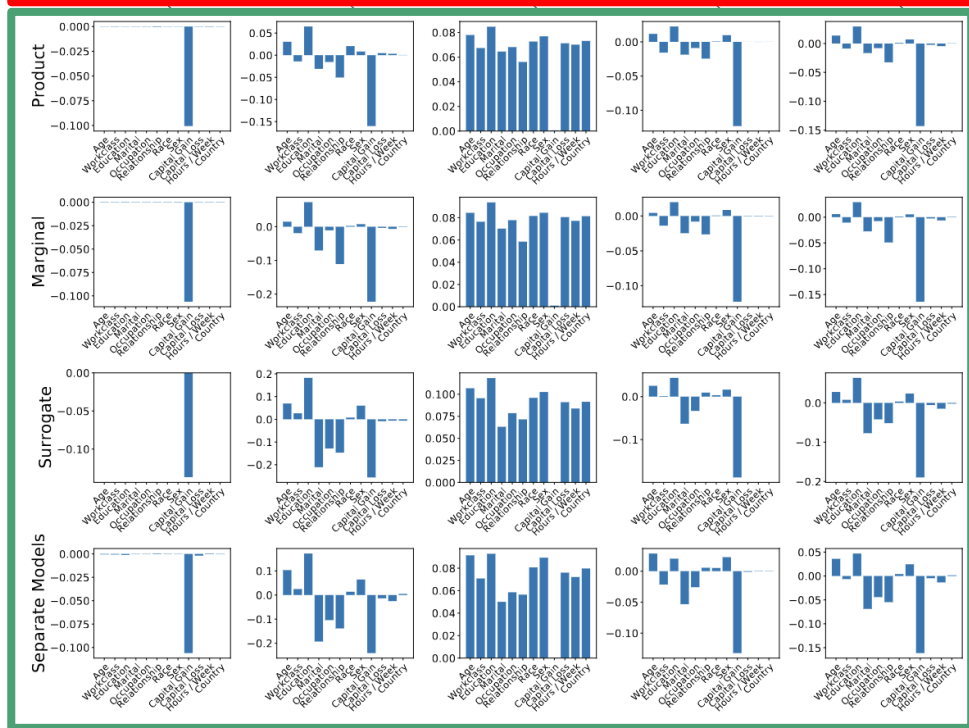
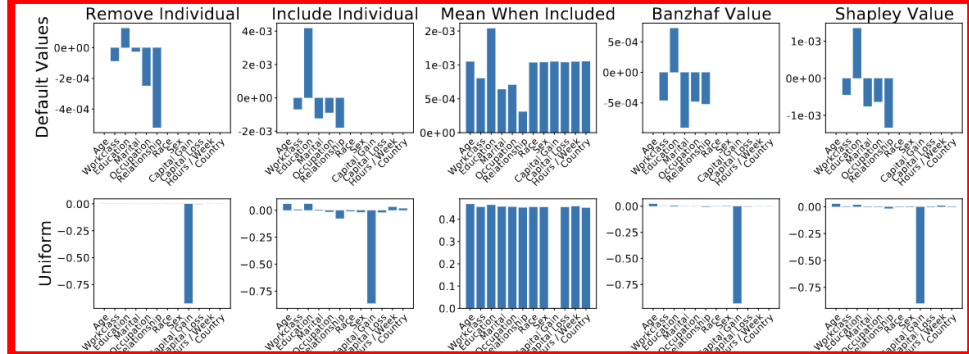
- 510 Patients
- Took a random subset of 100/17,814 genes
- Logistic Regression



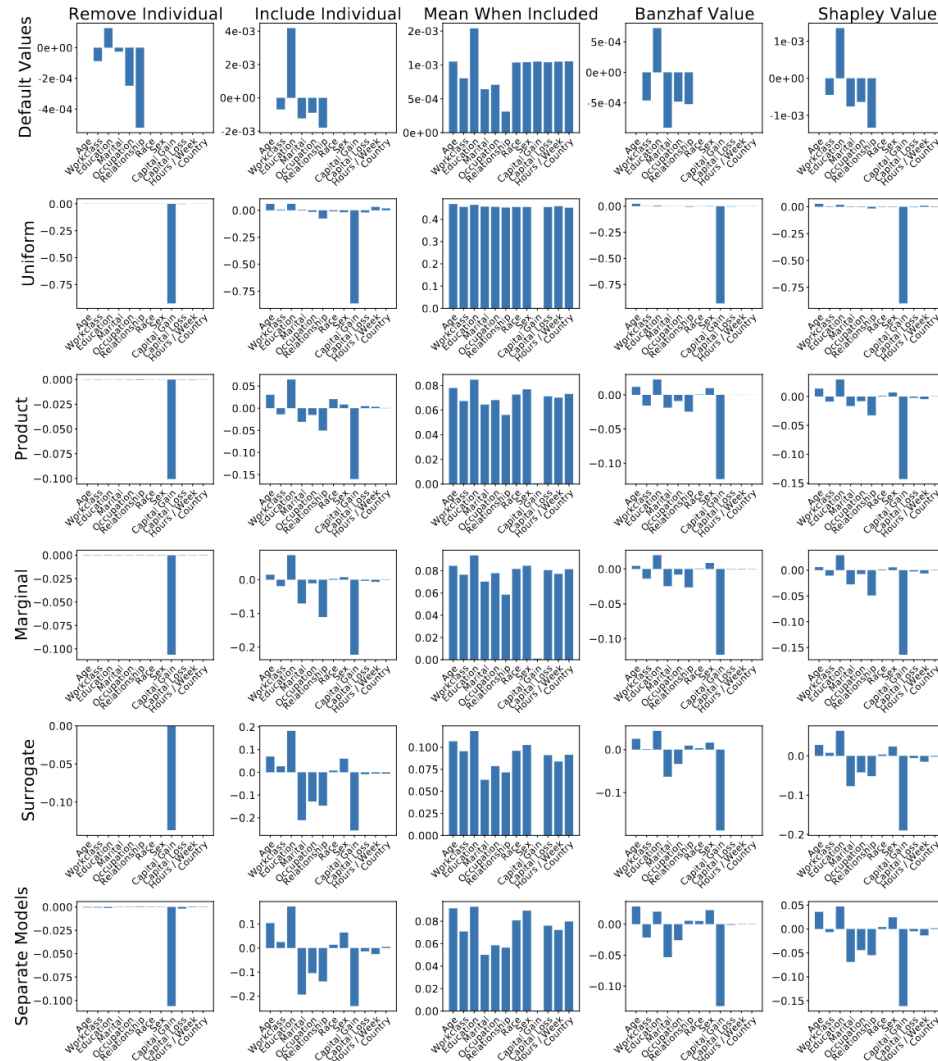
1. Census Income

Qualitatively:

- Bottom four the same
- Approximate conditional distribution



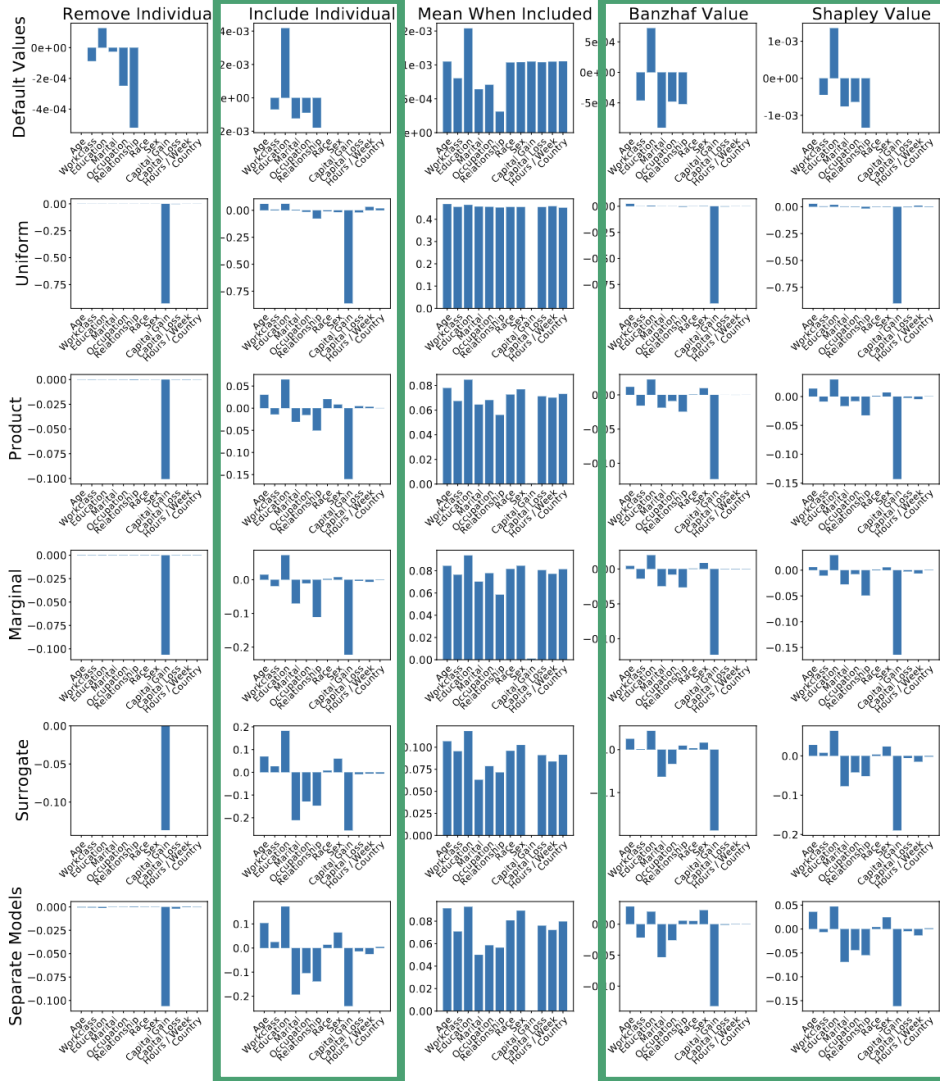
Qualitatively:



1. Census Income

Qualitatively:

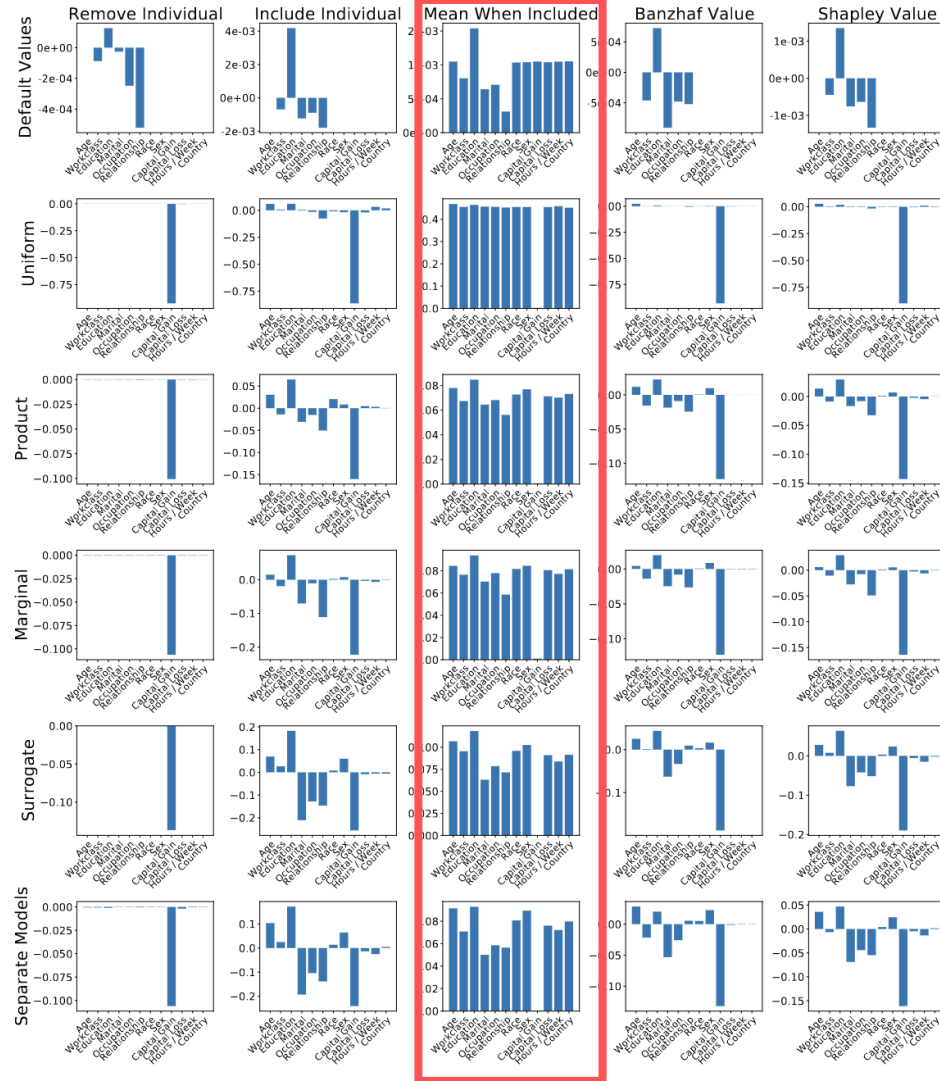
- Bottom four the same
 - Approximate conditional distribution
- Include and Banzhaf/Shapley the same
 - Similar formulations



1. Census Income

Qualitatively:

- Bottom four the same
 - Approximate conditional distribution
- Include and Banzhaf/Shapley the same
- Mean When Included is

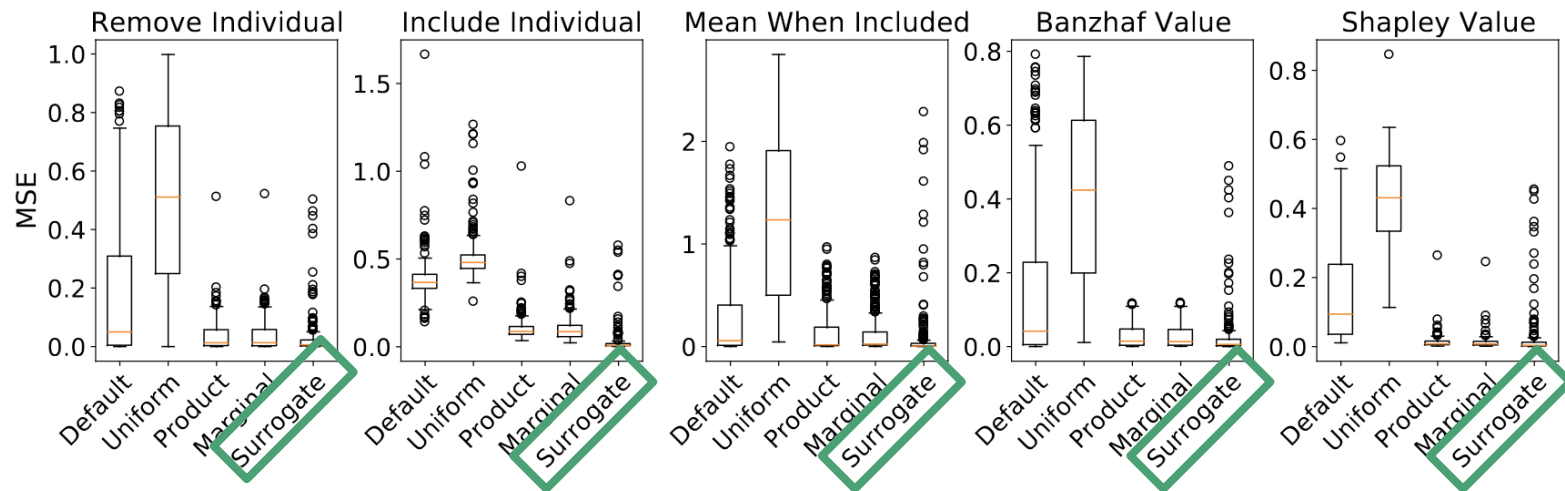
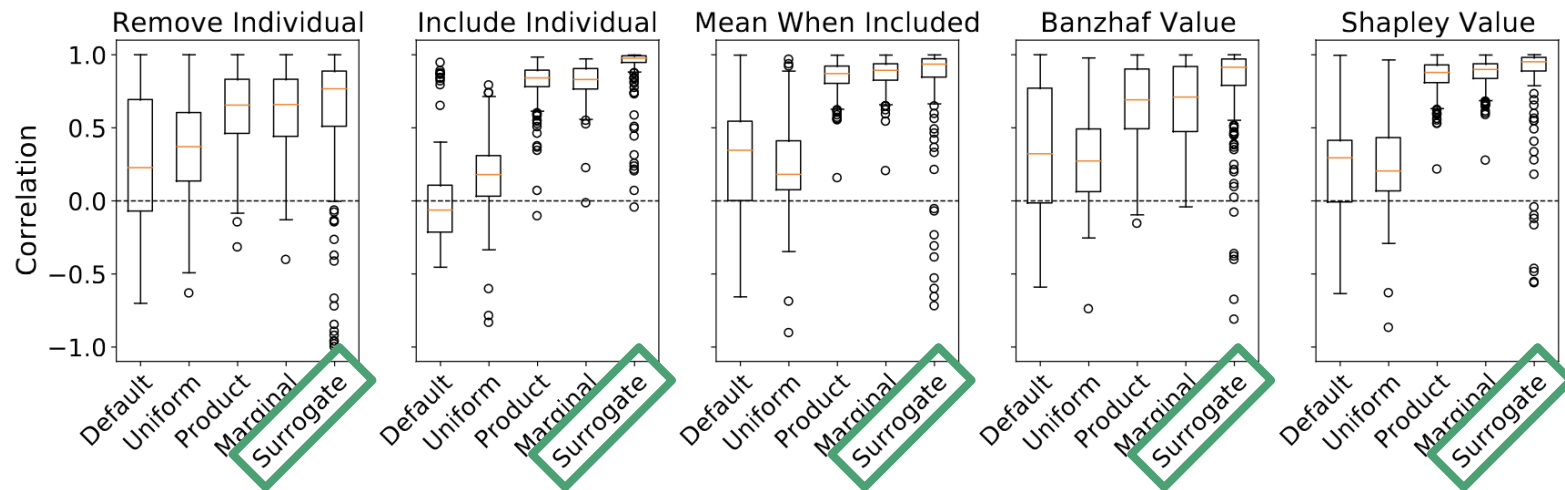


1. Census Income

Quantitatively:

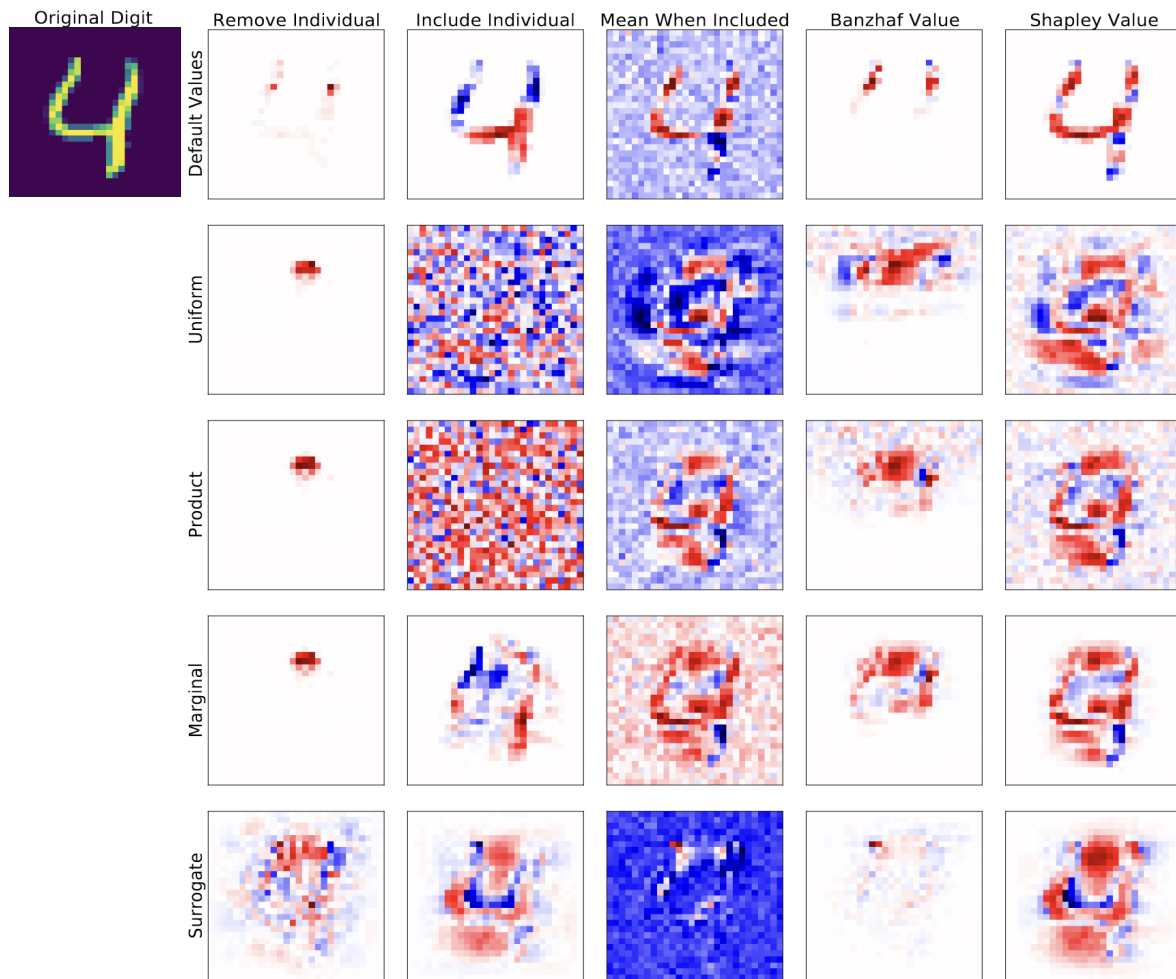
How good is the conditional distribution approximation?

- Intractable, so treat separate models as ground truth
- Surrogate Removal Method does best



2. MNIST

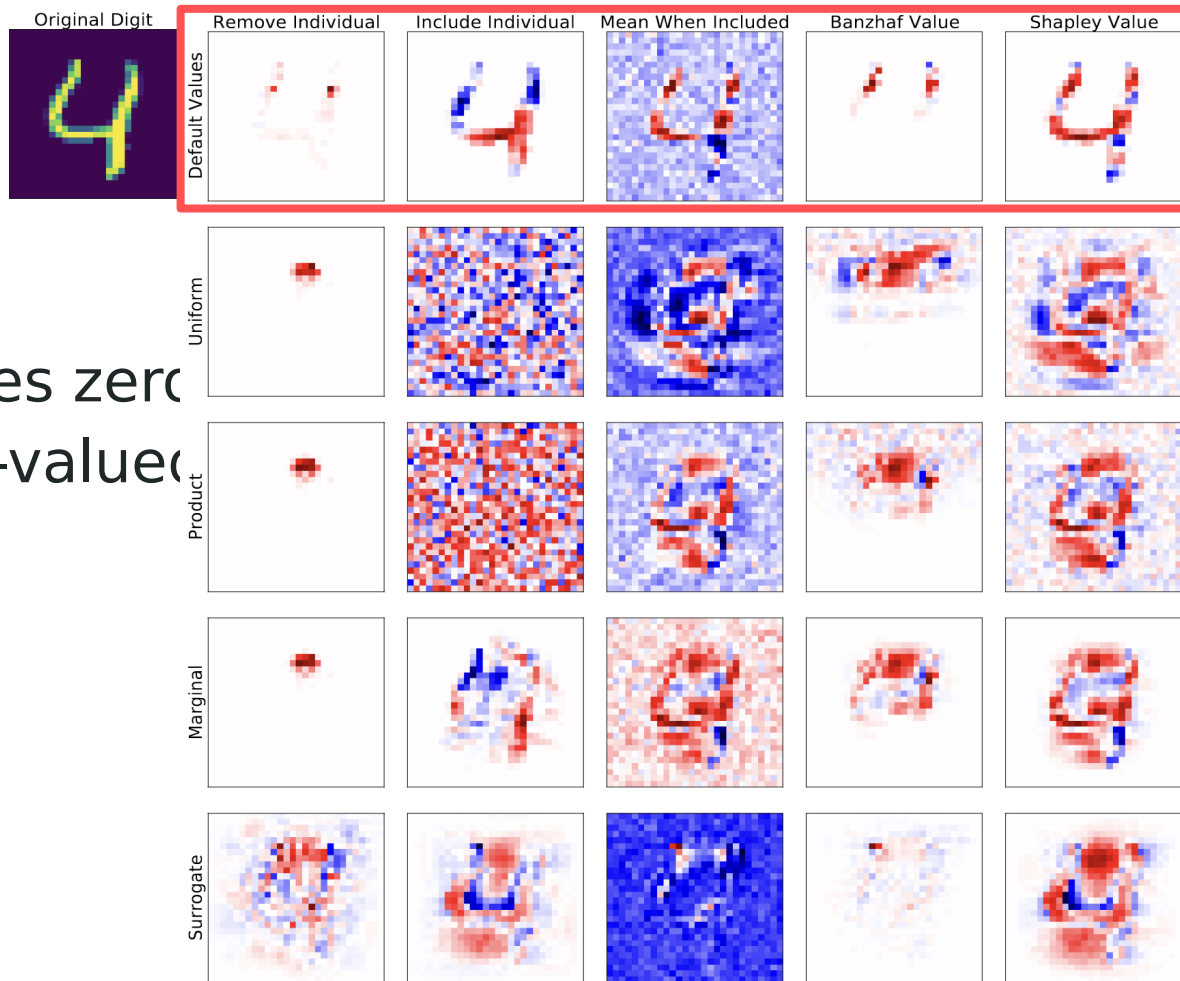
Qualitatively:



2. MNIST

Qualitatively:

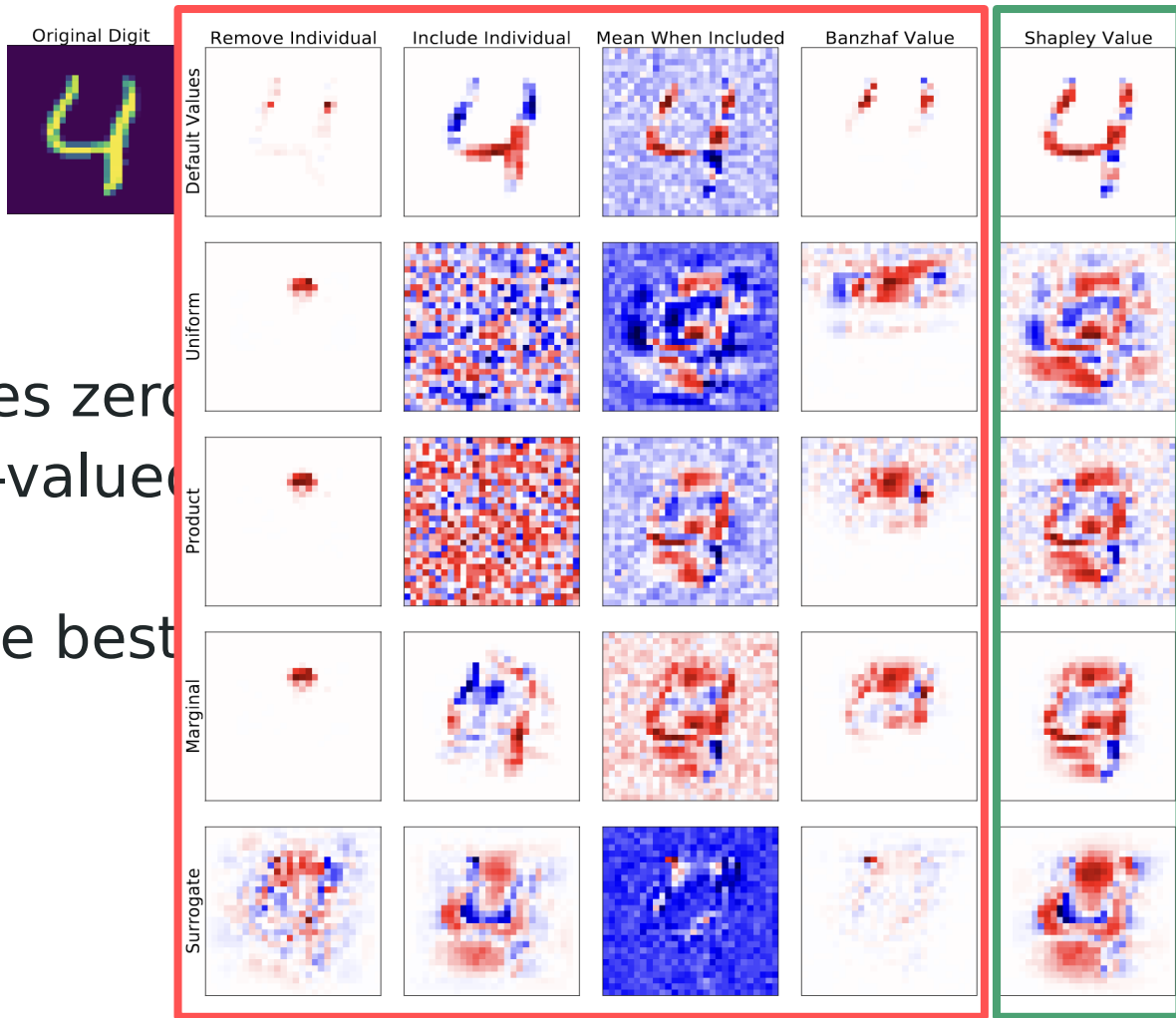
- Default Values gives zero attribution to zero-value features



2. MNIST

Qualitatively:

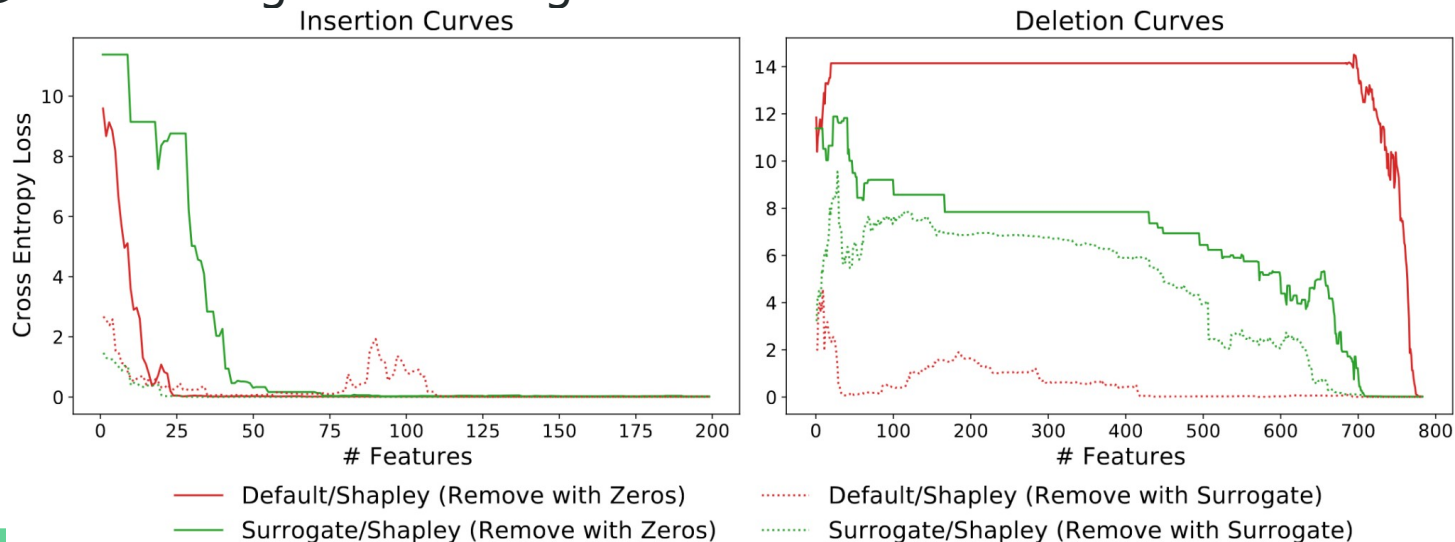
- Default Values gives zero attribution to zero-valued features
- Shapley “looks” the best



2. MNIST

● Quantitative Analysis:

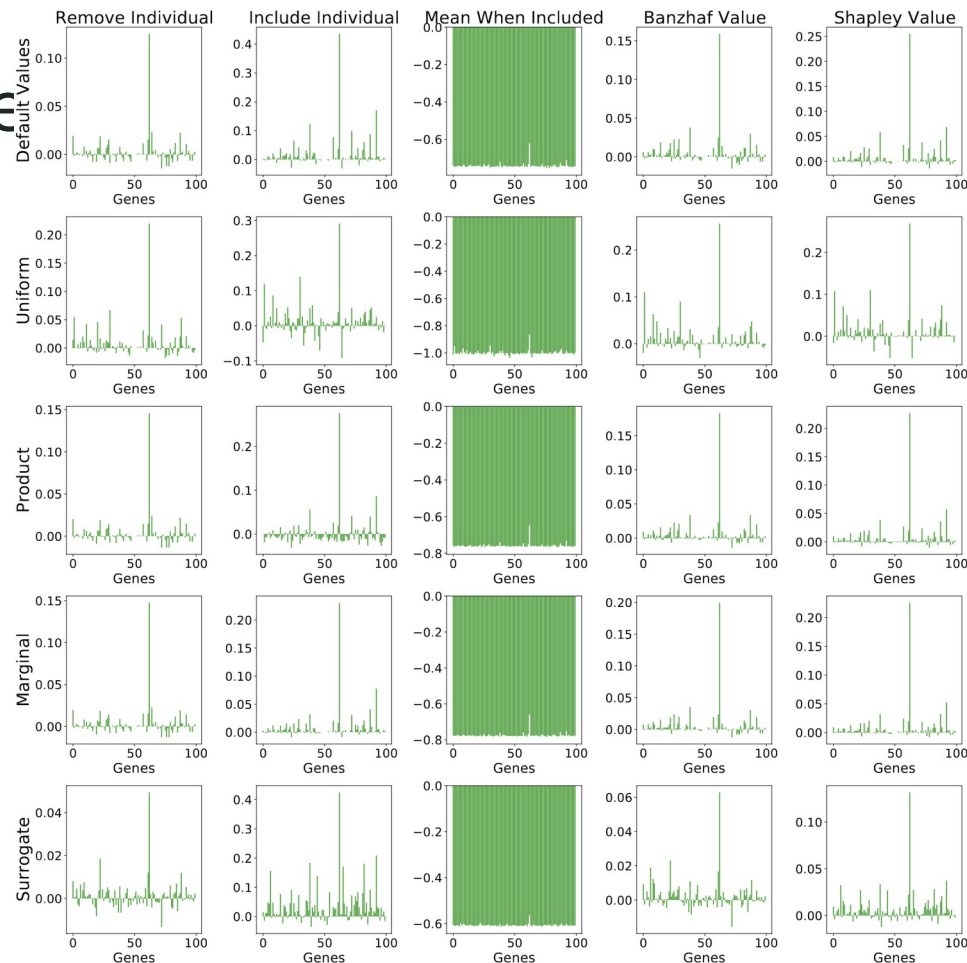
- Insertion: Remove the bottom k important features
- Deletion: Remove the top k important features
- Removing with surrogate is better!



3. Breast Cancer Subtype

Qualitatively:

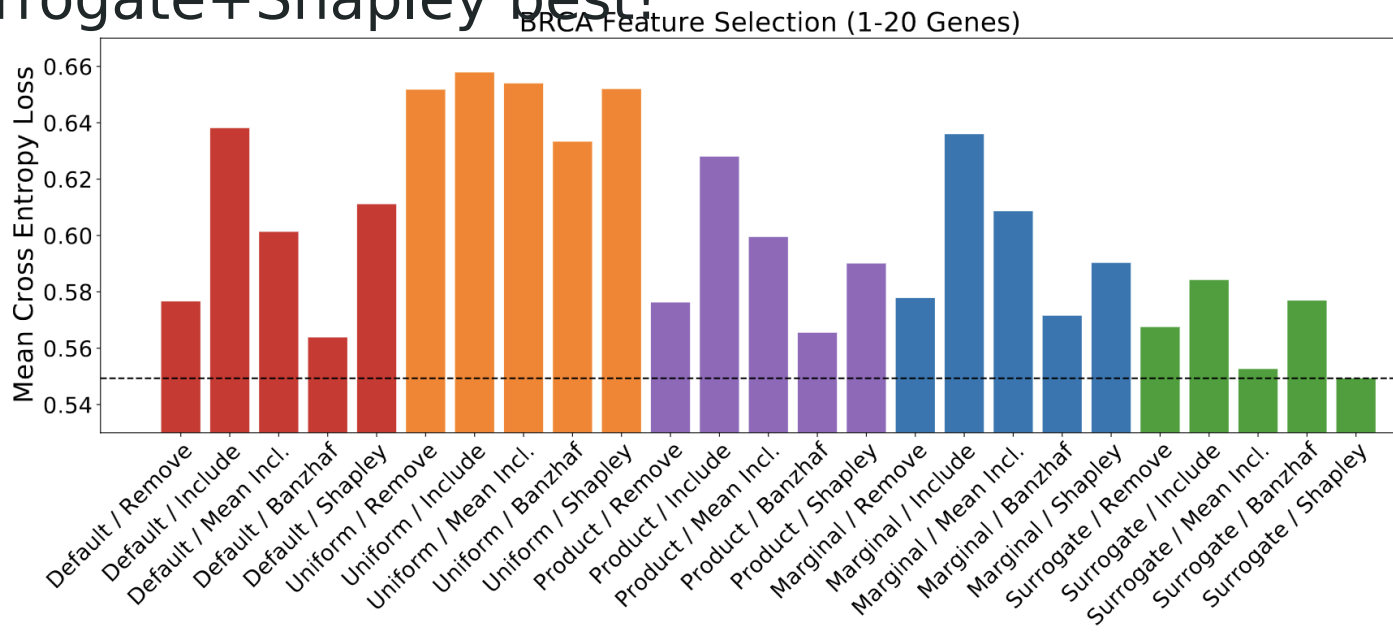
- ESR1 gene is most important
- Hard to compare qualitatively
- Verify that “mean when included” is much different from the rest



3. Breast Cancer Subtypes

Quantitatively:

- Select only 20 most important genes
- Surrogate+Shapley best!



Conclusion



- SHAP+Surrogate Model is the best
 - In line with information-theoretic connection
- The authors' unified framework helps us reason abstractly about these techniques
 - Able to identify and fill so many gaps in the combination of removal-based models

Limitations



- No limitations section
- No comparisons across model behavior
- Quantifying approximation quality with an approximation
- Evaluation metrics can be aligned with explanation method
 - Doesn't bode well for a universal unbiased metric
- Slightly suspect analysis for Breast Cancer dataset

Discussion

- Are removal-based the right framework to go in XAI?
- How do we reconcile with the fact that removing features may result in out-of-distribution behavior?
- What other unifying frameworks can you think of?
- Do you agree with the assessment that SHAP provides the best explanation?
- What methods do not fall under removal-based methods?

Which Explanation Should I Choose? A Function Approximation Perspective to Characterizing Post Hoc Explanations

Presented By:

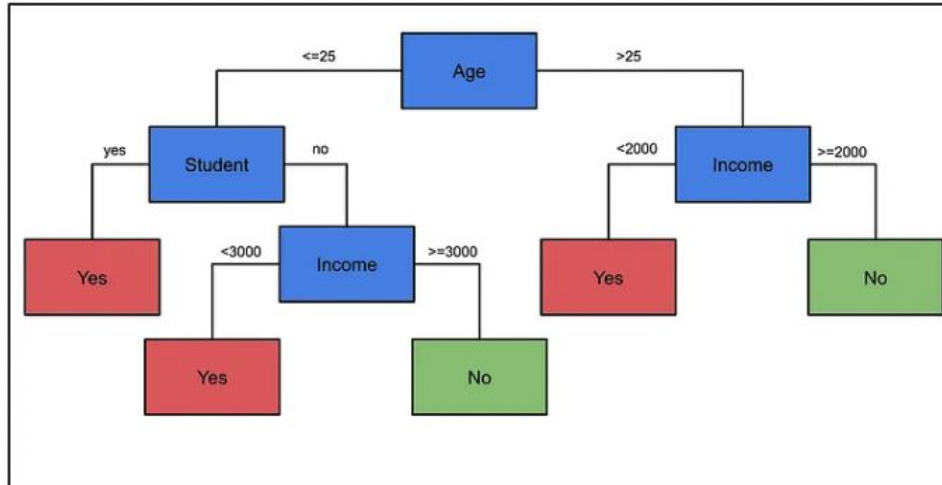
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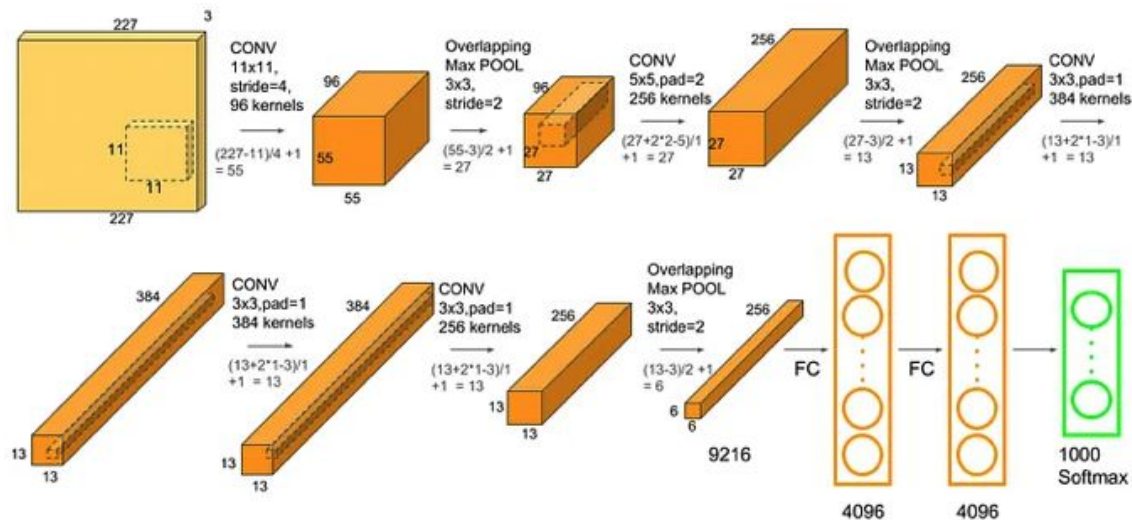
Interpretability

- Machine Learning models makes decision in high stake settings (healthcare, law, finance)
- Growing emphasis on understanding how models make predictions



Explainability

- Dealing with complicated models requires explainability through post hoc explanations



Post Hoc Explanation Methods

- LIME
- C-LIME
- SHAP
- Occlusion
- Vanilla Gradients
- Gradient x Input
- SmoothGrad
- Integrated Gradients

Different Methods Different Explanations

- Inconsistency in their goals
- What is an explanation?
- Lack of Mathematical Framework resulting in misunderstanding of their goals and properties

Summary of Key Contributions

- 1) Formalize the local function approximation (LFA) framework
- 2) Demonstrate that eight popular explanation methods can be characterized as instances of the LFA framework
- 3) No Free Lunch Theorem: no single LFA method can perform optimally across all noise neighbourhoods
- 4) Provide a guiding principle for choosing among LFA explanation methods based on the input domain

Related Work: Connections and Properties among post hoc explanation methods.

- C-LIME and SmoothGrad connections
- Gradient-based explanation methods and when they produce similar explanations.
- Faithfulness to the black-box model, robustness to adversarial attack, and fairness across subgroups.

Local Function Approximation Framework

Definition 1. *Local function approximation (LFA) of a black-box model f on a neighbourhood distribution $\mathcal{Z}$ around $\mathbf{x}_0$ by an interpretable model class $\mathcal{G}$ and a loss function ℓ is given by*

$$g^* = \arg \min_{g \in \mathcal{G}} \mathbb{E}_{\xi \sim \mathcal{Z}} \ell(f, g, \mathbf{x}_0, \xi) \quad (1)$$

where a valid loss ℓ is such that $\mathbb{E}_{\xi \sim \mathcal{Z}} \ell(f, g, \mathbf{x}_0, \xi) = 0 \iff f(\mathbf{x}_\xi) = g(\mathbf{x}_\xi) \quad \forall \xi \sim \mathcal{Z}$

Distinction with LIME

- The LFA framework requires that f and g share the same input domain X and output domain Y , suggesting LIME does not fall in LFA
- LFA framework ensures model recovery if $f \in G$ and $\text{domain}(x) = X$
- Optimization is done through splitting the perturbation data into train / validation / test sets. LIME does not, making it possible to overfit a small number of perturbations

Designing Explanations with LFA

LFA guides creation of explanations

LFA requires you define

1. interpretable model class $\mathbf{G}$
2. neighbourhood distribution $\mathbf{Z}$
3. loss function l
4. binary operator $\oplus$ to combine the input and the noise

Correspondence with others explanation methods

Explanation Method	Local Neighbourhood $\mathcal{Z}$ around $\mathbf{x}_0$	Loss Function ℓ
C-LIME	$\mathbf{x}_0 + \xi; \xi(\in \mathbb{R}^d) \sim \text{Normal}(0, \sigma^2)$	Squared Error
SmoothGrad	$\mathbf{x}_0 + \xi; \xi(\in \mathbb{R}^d) \sim \text{Normal}(0, \sigma^2)$	Gradient Matching
Vanilla Gradients	$\mathbf{x}_0 + \xi; \xi(\in \mathbb{R}^d) \sim \text{Normal}(0, \sigma^2), \sigma \rightarrow 0$	Gradient Matching
Integrated Gradients	$\xi \mathbf{x}_0; \xi(\in \mathbb{R}) \sim \text{Uniform}(0, 1)$	Gradient Matching
Gradients $\times$ Input	$\xi \mathbf{x}_0; \xi(\in \mathbb{R}) \sim \text{Uniform}(a, 1), a \rightarrow 1$	Gradient Matching
LIME	$\mathbf{x}_0 \odot \xi; \xi(\in \{0, 1\}^d) \sim \text{Exponential kernel}$	Squared Error
KernelSHAP	$\mathbf{x}_0 \odot \xi; \xi(\in \{0, 1\}^d) \sim \text{Shapley kernel}$	Squared Error
Occlusion	$\mathbf{x}_0 \odot \xi; \xi(\in \{0, 1\}^d) \sim \text{Random one-hot vectors}$	Squared Error

LFA with Continuous Noises: Gradient-Based Methods

Gradient Matching Loss:

$$\ell_{gm}(f, g, \mathbf{x}_0, \xi) = \|\nabla_{\xi} f(\mathbf{x}_0 \oplus \xi) - \nabla_{\xi} g(\mathbf{x}_0 \oplus \xi)\|_2^2$$

No Free Lunch Theorem for Explanation Methods

Theorem 3 (No Free Lunch for Explanation Methods). *Consider explaining a black-box model f around point $\mathbf{x}_0$ using an interpretable model g from model class $\mathcal{G}$ and a valid loss function ℓ where the distance between f and $\mathcal{G}$ is given by $d(f, \mathcal{G}) = \min_{g \in \mathcal{G}} \max_{\mathbf{x} \in \mathcal{X}} \ell(f, g, 0, \mathbf{x})$.*

Then, for any explanation g^ over a neighbourhood distribution $\xi_1 \sim \mathcal{Z}_1$ such that $\max_{\xi_1} \ell(f, g^*, \mathbf{x}_0, \xi_1) \leq \epsilon$, there always exists another neighbourhood $\xi_2 \sim \mathcal{Z}_2$ such that $\max_{\xi_2} \ell(f, g^*, \mathbf{x}_0, \xi_2) \geq d(f, \mathcal{G})$.*

When $\mathbf{g}$ is less expressive than $\mathbf{f}$, no single explanation $\mathbf{g}^*$ can perform optimally across all neighborhoods

- Example: $\mathbf{f}$ is non-linear and $\mathbf{g}^*$ is linear

Characterizing Explanation Methods via Model Recovery

Definition 2 (Model Recovery: Guiding Principle). *Given an instance of the LFA framework with a black-box model f such that $f \in \mathcal{G}$ and a specific noise type (e.g., Gaussian, Uniform), an explanation method performs model recovery if there exists some noise distribution $\mathcal{Z}$ such that LFA returns $g^* = f$.*

How can you evaluate whether an explanation $\mathbf{g}$ works for $\mathbf{f}$?

If $\mathbf{f}$ and $\mathbf{g}^*$ are the same model class, it should be possible for $\mathbf{g}$ to approximate ("recover") $\mathbf{f}$

Characterizing Explanation Methods via Model Recovery

Let's explore which of the 8 explanation work for various input domains $\mathbf{X}$. Three cases:

1. continuous $\mathbf{X}$
2. binary $\mathbf{X}$
3. discrete $\mathbf{X}$

1. Which Explanation for continuous X ? (1/2)

Assume f and g are linear ($f(\mathbf{x}) = \mathbf{w}_f^\top \mathbf{x}$ and $g(\mathbf{x}) = \mathbf{w}_g^\top \mathbf{x}$)

A)  Additive continuous noise (SmoothGrad, Vanilla Gradients, C-LIME). Why? $\mathbf{w}_g = \mathbf{w}_f$

$$\ell(f, g, \mathbf{x}_0, \xi) = \|\nabla_\xi f(\mathbf{x}_\xi) - \nabla_\xi g(\mathbf{x}_\xi)\|_2^2.$$

B)  Multiplicative continuous noise (Integrated Gradients and Gradient x Input). Why? loss function parameterization

$$\ell(\bar{f}, g, \mathbf{x}_0, \xi) = \|\nabla_\xi \bar{f}(\mathbf{x}_\xi) - \nabla_\xi g(\xi)\|_2^2.$$

1. Which Explanation for continuous X ? (2/2)

C)  Multiplicative binary noise (LIME, KernelSHAP, and Occlusion). Why? Consider sinusoidal example

Remark 2. For $\mathcal{X} = \mathbb{R}^d$, periodic functions f and g where $f(\mathbf{x}) = \sum_{i=1}^d \sin(\mathbf{w}_{f_i} \odot \mathbf{x}_i)$ and $g(\mathbf{x}) = \sum_{i=1}^d \sin(\mathbf{w}_{g_i} \odot \mathbf{x}_i)$, and an integer n , binary noise methods do not perform model recovery for $|\mathbf{w}_{f_i}| \geq \frac{n\pi}{\mathbf{x}_{0_i}}$.

$$\sin(\mathbf{w}_{f_i} \mathbf{x}_{0_i}) = \sin(\pm n\pi) = \sin(0) = 0$$

$\sin(\mathbf{w}_{f_i} \mathbf{x}_{0_i})$ outputs zero for all binary perturbations

discrete nature of noise makes model recovery impossible

2. Which Explanation for binary X ?

Consider binary noise methods (continuous noise invalid)

Only Multiplicative binary perturbations methods (LIME, KernelSHAP, and Occlusion) enable $\mathbf{g}$ to recover $\mathbf{f}$ in the binary domain

3. Which Explanation for discrete X?

-  continuous noise methods invalid
-  binary noise methods - same sinusoidal logic
- All 8 explanations fail
- Use LFA to design new explanation with a discrete noise type

Designing Explanations with LFA

LFA guides creation of explanations

LFA requires you define

1. interpretable model class **G**
2. neighbourhood distribution **Z**
3. loss function **l**
4. binary operator $\oplus$ to combine the input and the noise

Summary of Properties of Existing Explanation Methods

For continuous data

Method	Characteristics of ξ	g recovers f ?	Scale of g 's weights when $\mathcal{X} \in \mathbb{R}^d$
C-LIME	Continuous, Additive	When $\mathcal{X} \in \mathbb{R}^d$	Gradient
SmoothGrad	Continuous, Additive	When $\mathcal{X} \in \mathbb{R}^d$	Gradient
Vanilla Gradients	Continuous, Additive	When $\mathcal{X} \in \mathbb{R}^d$	Gradient
Integrated Gradients	Continuous, Multiplicative	No	Gradient $\times$ Input
Gradients $\times$ Input	Continuous, Multiplicative	No	Gradient $\times$ Input
LIME	Binary, Multiplicative	When $\mathcal{X} \in \{0, 1\}^d$	Gradient $\times$ Input
KernelSHAP	Binary, Multiplicative	When $\mathcal{X} \in \{0, 1\}^d$	Gradient $\times$ Input
Occlusion	Binary, Multiplicative	When $\mathcal{X} \in \{0, 1\}^d$	Gradient $\times$ Input

Table 2: Summary of properties of existing explanation methods in relation to the LFA framework. In this table, we consider the scale of g 's weights when $\mathcal{X} \in \mathbb{R}^d$.

For binary data

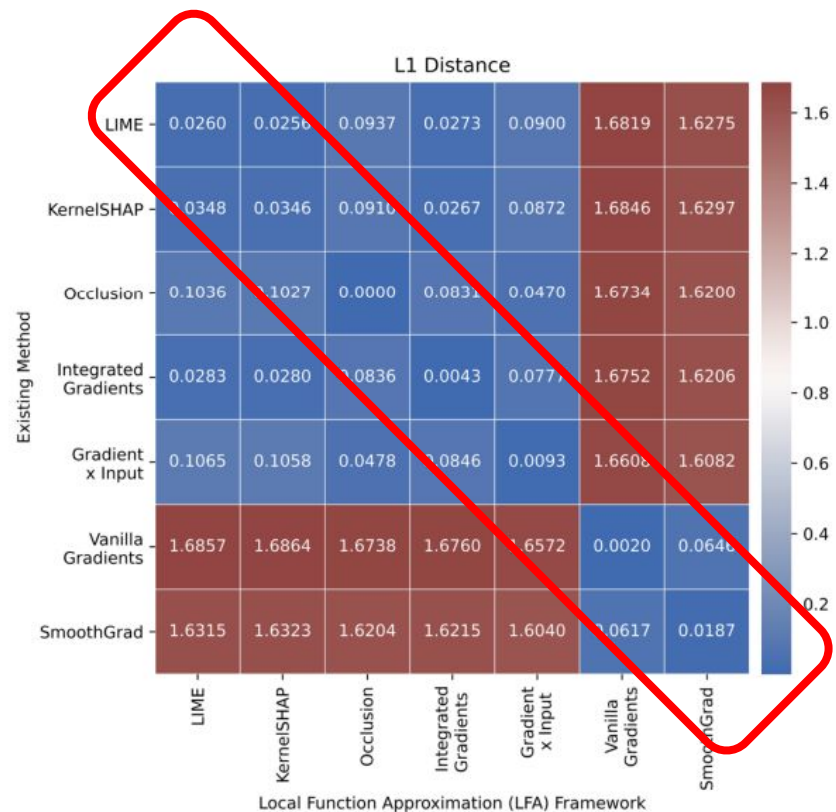
Empirical Evaluation - Dataset & Models

- 1) Common Experimental Goal: Examine different XAI methods' explanations of models' predictions on sample datasets to validate theoretical claims
- 2) Two Datasets
 - a) World Health Organization Life Expectancy dataset (20 features)
 - b) Home Equity Line of Credit (HELOC) dataset from FICO (24 features)
- 3) 4 Models:
 - a) 1 Simple Model (generalized linear regression)
 - b) 3 Neural Network Models of varying complexity

Empirical Evaluation - Experiment #1

- 1) **Goal:** Show each explanation method fits within the LFA framework by comparing the original method with a LFA re-implementation
- 2) **Methodology:**
 - a) For each XAI method (e.g. LIME), randomly select 100 test set points
 - b) Use the LFA and original (Meta) implementations to explain the predictions of black-box models
 - c) Evaluate the similarity of the two implementations' explanations (L1)
- 3) **Results:**
 - a) All 7 methods tested (excl. C-LIME) display near-zero L1 distances
 - b) Implies each method fits within the LFA framework

Empirical Evaluation - Experiment #1



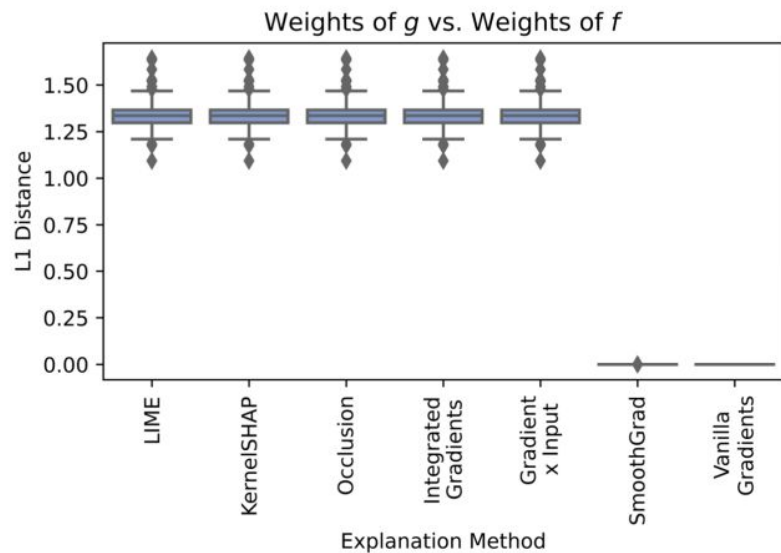
Empirical Evaluation - Experiment #2

- 1) **Goal:** Confirm that all XAI methods within the LFA achieve “model recovery”
 - a) Show that LFA methods recover the black box model “ f ” when f is an interpretable model

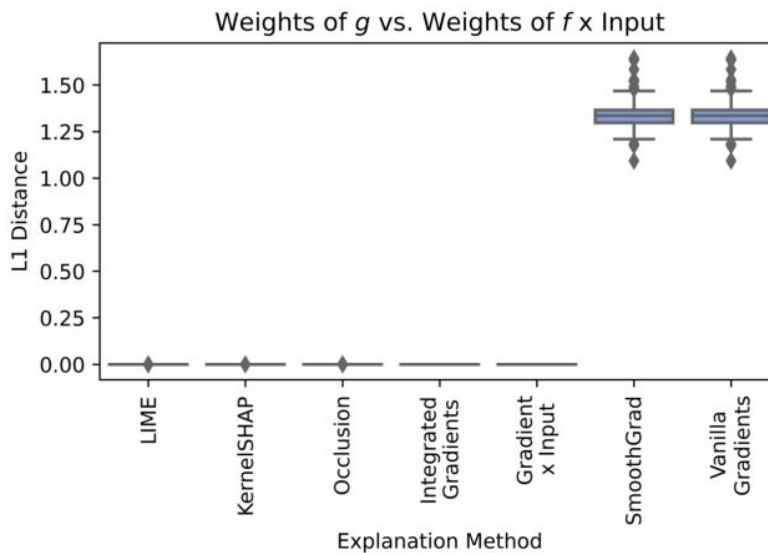
- 2) **Methodology:**
 - a) Set f to be the trained linear regression model on each dataset
 - b) Generate LFA explanations (g^*), where model class G is linear models
 - c) Compare weights of g^* with f

- 3) **Results:**
 - a) All 7 models satisfy the LFA guiding principle of “model recovery”
 - b) Weights of $g^* = (\text{gradient of } f) \text{ or } (\text{gradient of } f) \times (\text{input})$ for each method

Empirical Evaluation - Experiment #2



(a)



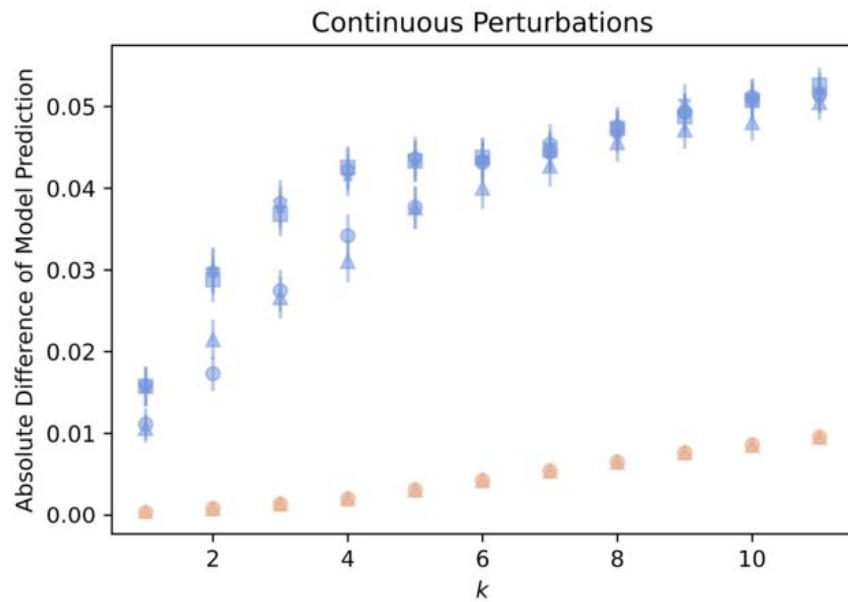
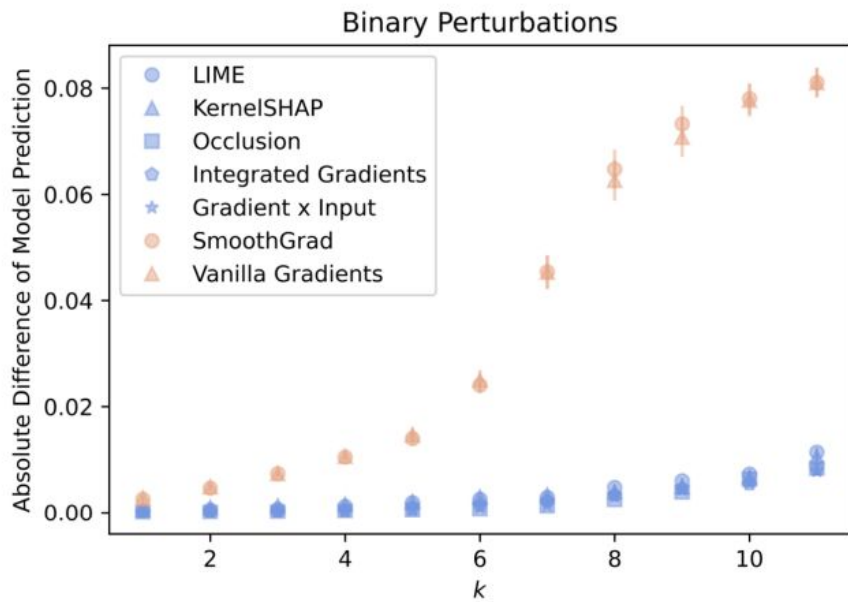
(b)

Empirical Evaluation - Experiment #3

- 1) **Goal:** Illustrate the LFA No Free Lunch Theorem for common XAI methods
- 2) **Methodology:**
 - a) For each method, generate explanations for 100 random test points
 - b) For $k = (1, 12)$, evaluate each explanation by
 - i) replace the top/bottom- k features with 0 (binary perturbation)
 - ii) adding Gaussian noise to the top/bottom- k features (continuous perturbation)
 - c) Calculate the absolute change in model prediction after perturbation
- 3) **Results:**
 - a) SmoothGrad & Vanilla Gradients perform less well for binary perturbations
 - b) 5 other methods perform less well for continuous perturbations

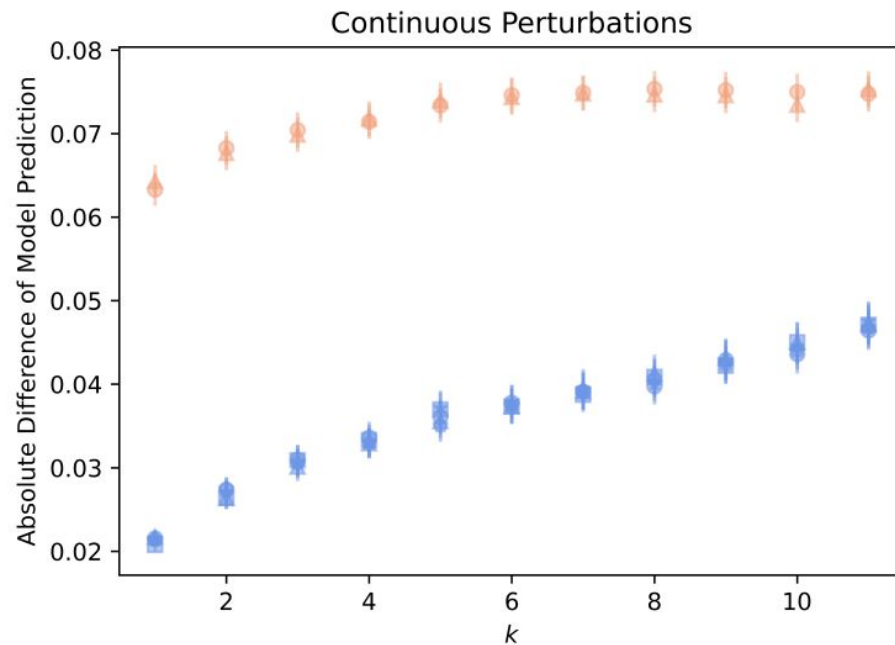
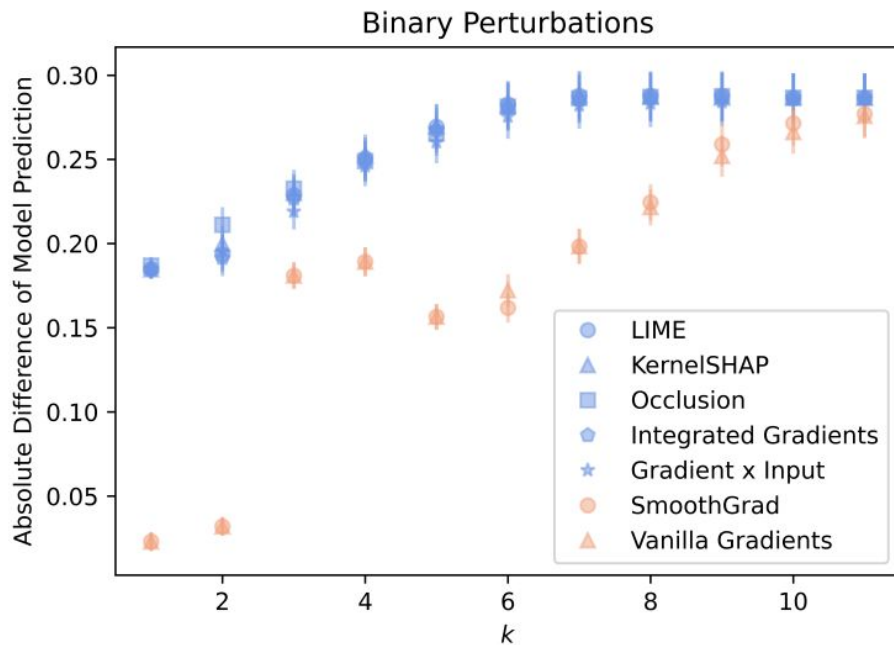
Empirical Evaluation - Experiment #3

Perturbation of Bottom-K Features



Empirical Evaluation - Experiment #3

Perturbation of Top-K Features



Summary of Key Contributions

- 1) Formalize the local function approximation (LFA) framework
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Future Work

- 1) Extend LFA analysis to additional post-hoc explanation methods
- 2) Develop a similar unifying conceptual framework for the *interpretability* of different model explanations
 - a) This work develops a unifying framework for evaluating *faithfulness* of explanations
 - b) May require more than a theoretical examination;
 - i) Human-Computer Interaction research such as user studies

Discussion Questions

- 1) Do you agree that the Local Function Approximation framework is a useful conceptual tool for understanding & comparing explanation methods?
- 2) If no explainability method can perform optimally across all perturbation distributions, as implied by the No Free Lunch Theorem, does that mean explainability should be defined relative to a perturbation neighborhood?
 - a) How does that complicate interpretability, especially for practitioners without ML expertise?
- 3) More broadly, what do you think about papers that attempt to create conceptual coherence & clarity across the field of explainability? Should this be a higher priority area of research?